



Securing understanding of arithmetic sequences

Task A: Reviewing the n^{th} term

1) Find the n^{th} term of these sequences.

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|----|--|--|--------------------------------|
| a) | 11, 19, 27, 35, 43, ... | Common difference: +8
Translation from $8n$: +3 | $8n + 3$ |
| b) | 4, 12, 20, 28, 36, ... | Common difference: +8
Translation from $8n$: -4 | $8n - 4$ |
| c) | 11, 7, 3, -1, -5, ... | Common difference: -4
Translation from $-4n$: +15 | $-4n + 15$ or... $15 - 4n$ |
| d) | 1.1, 1.9, 2.7, 3.5, 4.3, ... | Common difference: +0.8
Translation from $0.8n$: +0.3 | $0.8n + 0.3$ |
| e) | $\frac{4}{11}, \frac{7}{11}, \frac{10}{11}, \frac{13}{11}, \frac{16}{11}, \dots$ | Common difference: $+\frac{3}{11}$
Translation from $\frac{3}{11}n$: $+\frac{1}{11}$ | $\frac{3}{11}n + \frac{1}{11}$ |

Task B: Using the n^{th} term

1) For each sequence:

(i) find the n^{th} term,

(ii) find the 100th term,

(iii) identify if 500 is in the sequence and if so, its position.

- | | | |
|----|------------------------------|---|
| a) | -10, -7, -4, -1, 3, ... | <p>(i) n^{th} term: $3n - 13$</p> <p>(ii) 100th term: $3(100) - 13 = 287$</p> <p>(iii) $3n - 13 = 500$</p> $3n = 513$ $n = 171$ <p>500 is the 171st term.</p> |
| b) | 999, 996, 993, 990, 987, ... | <p>(i) n^{th} term: $1002 - 3n$</p> <p>(ii) 100th term: $1002 - 3(100) = 702$</p> <p>(iii) $1002 - 3n = 500$</p> $502 = 3n$ $n = 167.3 \text{ (to 1d.p.)}$ <p>500 is not a term in the sequence.</p> |
| c) | 8.4, 8.8, 9.2, 9.6, ... | <p>(i) n^{th} term: $0.4n + 8$</p> <p>(ii) 100th term: $0.4(100) + 8 = 48$</p> <p>(iii) $0.4n + 8 = 500$</p> $0.4n = 492$ <p>500 is the 1230th term</p> |

c) 8.4, 8.8, 9.2, 9.6, ...

(i) n^{th} term: $0.4n + 8$

(ii) 100^{th} term: $0.4(100) + 8 = 48$

(iii) $0.4n + 8 = 500$

$0.4n = 492$

$n = 1230$ 500 is the 1230^{th} term.

2) Assuming each pattern continues to grow the same way, will we see one with exactly 250 circles?



5, 8, 11, ...

$3n + 2 = 250$

$3n + 2$

$3n = 248$

No

$n = 82.6$



8, 10, 12, ...

$2n + 6 = 250$

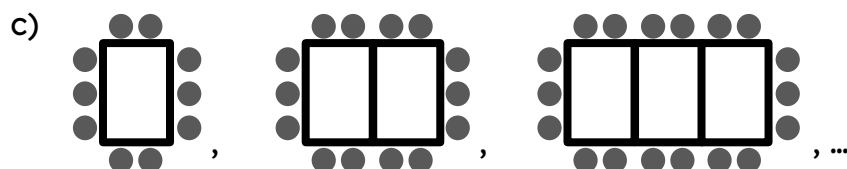
$2n + 6$

$2n = 244$

$n = 122$

Yes

(All evens ≥ 8)



10, 14, 18, ...

$4n + 6 = 250$

$4n + 6$

$4n = 244$

$n = 61$

Yes

Task C: Generalising finding the n^{th} term

1) Write the first five terms of these sequences:

a) $5 + 4(n - 1)$

$a = 5, d = 4$

5, 9, 13, 17, 21, ...

b) $4 + 5(n - 1)$

$a = 4, d = 5$

4, 9, 14, 19, 24, ...

c) $4 - 5(n - 1)$

$a = 4, d = -5$

4, -1, -6, -11, -16, ...

Hint: They are written in the form $a + d(n - 1)$ where a is the first term and d is the common difference.



2) Write two different generalisations for each of these arithmetic sequences:

a) 7, 10, 13, 16, 19, ...

$$3n + 4$$

$$7 + 3(n - 1)$$

b) 7, -3, -13, -23, -33, ...

$$17 - 10n$$

$$7 - 10(n - 1)$$

c) 0.7, 0.95, 1.2, 1.45, 1.7, ...

$$0.25n + 0.45$$

$$0.7 + 0.25(n - 1)$$

d) $\frac{1}{7}, \frac{5}{7}, \frac{9}{7}, \frac{13}{7}, \frac{17}{7}, \dots$

$$\frac{4}{7}n - \frac{3}{7}$$

$$\frac{1}{7} + \frac{4}{7}(n - 1)$$

3) For the sequence $24 + 7(n - 1)$

a) Find the first five terms.

$$24, 31, 38, 45, 52, \dots$$

b) Find the 100th term.

$$24 + 7(100 - 1) = 24 + 7(99) = 717$$

c) Find the position of the term 388 in the sequence.

$$24 + 7(n - 1) = 388$$

$$7(n - 1) = 364$$

$$n - 1 = 52$$

$$n = 53$$

